

BASSAM WALID

Mechatronics Engineer | Control Systems, Automation & Embedded Systems

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SUMMARY

Mechatronics Engineering graduate from the German University in Cairo (High Honours, full scholarship) with hands-on experience in control systems, embedded hardware and PLC automation. Built an sEMG-controlled prosthetic hand for an A+ bachelor thesis, reaching 98.25% finger-flexion detection, and deployed nonlinear and path-tracking controllers on real hardware. Seeking a graduate engineering role in automation, robotics or control.

EDUCATION

Bachelor of Science (BSc) in Mechatronics Engineering

2021 - Jul 2026

German University in Cairo (GUC), Cairo, Egypt

- GPA 1.16 on the German grading scale (lower is better); High Honours; GUC Certificate of Appreciation.
- Full academic scholarship; ranked in the top 10 of ~160 students in two semesters (2023 and 2023/2024).
- Relevant coursework: Reinforcement Learning & Optimal Control (LQR, MPC), bond-graph modeling, image processing, probabilistic robot localization.

IGCSE Diploma

2021

New Ramses College, Cairo, Egypt

- Achieved a top nationwide IGCSE score.

ENGINEERING PROJECTS

sEMG-Controlled Tendon-Driven Prosthetic Hand — Bachelor Thesis, Grade A+ | May 2025

Official title: Brain-Computer Interface for Prosthetic Device Using EMG

- Designed and iterated the hand in SolidWorks, then 3D-printed it in PLA and TPU for ~5,000 EGP (~US\$100).
- Developed real-time surface electromyography (sEMG) control on an STM32F401: 10 ms sampling, EMA filtering and an adaptive threshold driving N20 DC motors and an MG90S servo.
- Reached **100%** grip/wrist-flexion detection and **98.25%** individual finger-flexion detection.
- Carried 5 kg (~49 N) without holding torque, with up to 95° finger rotation and a calculated 0.835 s flexion time.
- In a separate course project, modeled the hand's dynamics with bond graphs in 20-sim (sEMG chain, DC motors, gearing, tendon compliance), tuning against 40 N tendon-force and 45° flexion targets.

Furuta Rotary Inverted Pendulum — Nonlinear Control | MATLAB/Simulink, Raspberry Pi 5

- Derived the nonlinear dynamics with the Euler–Lagrange formulation and built the model in MATLAB/Simulink.
- Implemented a backstepping controller to stabilize the pendulum, validated in real time on a Raspberry Pi 5.

Autonomous Vehicle Path Tracking — Pure Pursuit | ROS, Gazebo, Raspberry Pi, team project

- Deployed a pure pursuit controller on a Raspberry Pi; the vehicle followed the requested path in ROS/Gazebo simulation and on physical hardware.

PLC Logic and Electro-Pneumatic Control — Mechatronics Lab | FluidSIM

- Programmed a logic controller in function block diagram (FBD), using AND/OR logic and set/reset latches to drive a motor, horns, an indicator lamp and a solenoid through relays from push-button and switch inputs.
- Built an electro-pneumatic circuit that sequences two double-acting cylinders using 5/2 double-solenoid valves and limit switches.

Wall-Climbing RC Robot | STM32 Blue Pill, PID, gyroscope

- Engineered an STM32-based robot whose gyroscope-driven PID control adjusts propeller direction to counter gravity, holding position or driving on walls of varying inclination.

Additional Projects

- Robotic pick-and-place gripper:** four servos, modeled in MATLAB/Simulink and controlled by a Raspberry Pi.
- IoT alarm clock (personal project):** ESP32 with Telegram bot control, RTC, I2C LCD, SD card and a weather API.
- FPGA traffic light controller:** four-way intersection state machine in VHDL on a Basys 3 FPGA.

TECHNICAL SKILLS

Control & Modeling: PID, backstepping, sliding mode control (SMC), fuzzy logic, LQR, MPC, Euler–Lagrange modeling, bond graphs, stability analysis

Automation: PLC programming (FBD), relay control circuits, electro-pneumatics (5/2 solenoid valves, limit switches)

Embedded & Electronics: STM32, ESP32, Raspberry Pi, Arduino, Basys 3 FPGA; I2C, SPI, UART, CAN, PWM; motor drivers, power electronics, sensor integration

Software & Simulation: MATLAB/Simulink, Simscape, 20-sim, FluidSIM, ROS, Gazebo, OpenCV, STM32CubeIDE

Programming: C, C++, Python, MATLAB, VHDL, Java; Git, Linux

CAD & Tools: SolidWorks, Fusion 360, AutoCAD, 3D printing (PLA, TPU), Microsoft Office (Word, Excel, PowerPoint)

EXPERIENCE

IGCSE Mathematics & Physics Tutor (Part-time)

Sep 2021 - Present

Self-employed, Cairo

- Tutored over 100 students alongside full-time engineering studies; **~76%** achieved A* or A.

IGCSE Physics Teacher

Apr 2026 - May 2026

Seeds British School, Cairo

- Taught IGCSE Physics classes, emphasizing conceptual understanding and problem solving.

CAD Design Intern

Jan 2025 - Feb 2025

PIMC, German University in Cairo

- Completed hands-on training in 3D CAD modeling with Autodesk Fusion 360.

Business Development & Customer Experience Intern

Aug 2023 - Oct 2023

Venu, Cairo

- Researched and onboarded new venues through outreach, and supported customer onboarding and bookings.

LEADERSHIP & ACTIVITIES

Speaker Coach

2023

TEDxGUC, German University in Cairo

- Coached two speakers through a joint TEDx talk from script to stage, helping write the script and training their delivery, body language and emphasis; the talk is published on YouTube.

ECOSOC Coordinator and Delegate

Model United Nations (MUN): coordinator for 1 year, delegate for 2 years

- Delivered training sessions to delegates as coordinator of the Economic and Social Council (ECOSOC).
- Won Best Delegate** among 200+ delegates at a national MUN competition at Misr International University.

LANGUAGES

Arabic (Native) | **English** (Fluent; IELTS 8.0, C1) | **German** (A2)